Juan David Muñoz-Bolaños
Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

March 19, 2026

JENOPTIK AG (SwissOptic)
Frau Edith Memeti
Heinrich-Wild-Strasse
9435 Heerbrugg
Schweiz

Subject: Application as Process Engineer Metrology 100% (m/f/d)

Dear Ms. Memeti,

As a PhD physicist with deep expertise in optical metrology, I am applying for the position of Process Engineer at your Heerbrugg site. I am specifically seeking a role where I can apply my analytical skills to ensure your manufacturing processes operate at the highest level of precision.

During my PhD, my focus was on building and validating high-resolution optical measurement systems (interferometry, wavefront analysis). To ensure high process stability, I use LabVIEW and Python extensively for the automation of calibration procedures. Previously, at INTECOL S.A.S., I was responsible for the design of industrial inspection systems in compliance with industry standards (e.g., ISO 10110).

Fully integrated into the DACH work culture, I will be available for employment from May 2026. I look forward to sustainably optimizing development and qualification processes at SwissOptic/Jenoptik.

Sincerely,

Juan David Muñoz-Bolaños
Prozessingenieur Messtechnologie
Medizinische Universität Innsbruck

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Prozessingenieur Messtechnologie

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PROFESSIONAL SUMMARY

Physicist and Optical Engineer with extensive expertise in precision metrology, interferometry, and high-resolution instrument development. Proven track record in bringing complex optical systems from concept to industrial-grade prototypes. Strong foundation in Python/C++ automation, ZEMAX simulation, and quality assurance under industrial norms (ISO 10110/9001).

TECHNICAL COMPETENCIES

Optical Metrology & System Design
End-to-end design and assembly of multiphoton/confocal microscopes and interferometric setups. Expertise in wavefront sensing and diffraction-limited system alignment.

Simulation & Tolerancing
Advanced use of ZEMAX OpticStudio for systematic tolerance analysis, sensitivity characterization of custom objectives, and Zernike-based wave-field simulations in Python.

Software & Automation
Expert proficiency in C++, Python (NumPy, Pandas, Scikit-learn, JAX), and LabVIEW/FPGA for real-time hardware control, detector calibration, and automation of calibration routines.

Industrial Vision & QA
Hands-on experience with Halcon and Cognex VisionPro for production-line inspection. Proficient in MSA (Gage R&R) and designing custom illumination concepts (Bright-field/Dark-field).

Standards & Compliance
Strong awareness of ISO 10110 and ISO 9001 standards. Experienced in managing EH&S compliance and maintaining technical documentation for high-power laser environments.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher - Optical Metrology & Automation — *Jan 2022 – Present*

- **Metrology System Build:** Designed and automated a custom wavefront sensing and phase retrieval system from scratch; wrote complete control stacks in C++/Python to interface cameras with deformable mirrors.
- **Optical Characterization:** Performed rigorous ZEMAX tolerance analysis to characterize aberration sensitivity in custom-built microscope objectives, validated via experimental interferometry.
- **R&D Project Leadership:** Led the hardware-software integration of a multiphoton microscope system, achieving diffraction-limited performance while managing vendor qualifications (Thorlabs, Edmund Optics).
- **Lab Operations:** Primary operator for a high-power laser lab, managing safety compliance (EH&S), equipment maintenance logs, and technical spec-sheet documentation.

Leibniz Institute of Photonic Technology (IPHT) — Jena, Germany
R&D Researcher (Photonics & Data) — *Jun 2020 – Dec 2021*

- **Spectrometer Prototyping:** Designed and built a 1064 nm fiber-probe Raman spectrometer for non-invasive clinical use; handled end-to-end optical layout, dispersive optic alignment, and detector calibration.
- **Data Science & ML:** Applied supervised Machine Learning to hyperspectral photonic data using custom Python packages, achieving high-accuracy tissue classification and accelerating experimental validation

loops.
- **Interdisciplinary Research:** Collaborated on clinical spectroscopy projects, resulting in two peer-reviewed publications in Applied Sciences and ACS Photonics.

INTECOL S.A.S. Medellín, Colombia
Junior Machine Vision System Engineer *Mar 2018 – Mar 2019*

- **Industrial Vision Pipelines:** Engineered Halcon pipelines for fast-moving production lines (glass bottle inspection), implementing custom illumination concepts to solve liquid level and label printing quality issues.
- **System Commissioning:** Successfully deployed automated inspection solutions for clients like Renault, utilizing Cognex VisionPro for robotic part recognition and flame-surface treatment.
- **Process Reliability:** Conducted feasibility studies and Gage R&R assessments to ensure measurement traceability and process stability under industrial conditions.

EDUCATION

Medical University of Innsbruck Austria
Ph.D. in Adaptive Optics & Microscopy *Jan 2022 – Present*

Friedrich Schiller University Jena Germany
M.Sc. in Photonics *2019 – 2021*

University of Cauca Colombia
B.Sc. in Physics Engineering *2012 – 2018*

PUBLICATIONS & AWARDS

- **Awards:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for top Colombian professionals.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte Medical University of Innsbruck
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft Leibniz Institute of Photonic Technology
Group leader: Spectroscopy Imaging
christoph.kraft@leibniz-ipht.de

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LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

AGREEMENT FOR GUESTS

to carry out a practical training

with

Mr. **Juan David Munoz Bolanos** **Date of birth: 1994-06-04**

temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** **(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolanos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 13 November 2020

F. Sondermann

Prof. J. Popp
Chairman
of Leibniz-IPHT

F. Sondermann
Executive Member
of Leibniz-IPHT

Juan David Munoz Bolanos
Guest

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LEIBNIZ-INSTITUT *für* PHOTONISCHE TECHNOLOGIEN

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Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

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Jena, 18 May 2020

F. Sondermann
Juan D. M.

Prof. J. Popp Chairman of Leibniz-IPHT	F. Sondermann Executive Member of Leibniz-IPHT	Juan David Munoz Bolanos Guest

INTECOL
OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos ($1,500,000) plus three hundred thousand pesos ($300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL
NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO
Administrative and Financial Director

FRIEDRICH-SCHILLER-UNIVERSITÄT JENA
PHYSIKALISCH-ASTRONOMISCHE FAKULTÄT

URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

Prof. Dr. Christian Spielmann
Dekan

Prof. Dr. Silvana Botti
Vorsitzende des Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20 DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC OF COLOMBIA.

UNIVERSIDAD DEL CAUCA

ON BEHALF OF THE

REPUBLIC OF COLOMBIA

AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION

CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS

CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,

GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396

RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY

THE DEAN OF THE FACULTY

THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR E INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINJUSTICIA 1997

CERTIFICATE

Juan Munoz Bolanos

has participated in the

ZEISS European Summer School

Lithography Optics & Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Dr. Thomas Marschner
Program Manager Funding Projects

Dr. Thomas Stammler
CTO & Head of Product Strategy

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden

österreich ösd INNSBRUCK schweiz deutschland

Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

Europäisches Siegel für innovative Spracheninitiativen 1998

ÖSD ist Mitglied von

ALTE

ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum
BFI Tirol Bildungs GmbH
in Innsbruck / Österreich

bestanden

österreich ösd INNSBRUCK schweiz deutschland

Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

Europäisches Siegel für innovative Spracheninitiativen 1998

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ösd
österreich schweiz deutschland

ID-Nummer: ZB2Ö25m40434

REPUBLICA DE COLOMBIA

PASAPORTE
PASSPORT

Tipo / *Type*: P
Cod. país / *Country code*: COL
Pasaporte Nº / *Passport No.*: AV410884

Apellidos / *Surname*
MUÑOZ BOLAÑOS

Nombres / *Given names*
JUAN DAVID

Nacionalidad / *Nationality*
COLOMBIANA

Fecha de nacimiento / *Date of birth*: 04 JUN/JUN 1994
Núm. personal / *Personal No.*: CC1061772278

Sexo / *Sex*: M
Lugar de nacimiento / *Place of birth*: LA CRUZ COL

Fecha de expedición / *Date of issue*: 11 DIC/DEC 2018
Autoridad / *Authority*: G. ANTIOQUIA

Fecha de vencimiento / *Date of expiry*: 10 DIC/DEC 2028
Firma del titular / *Holder's signature*

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UNIVERSITÄT INNSBRUCK
AUSSTELLUNGSDATUM / DATE OF ISSUE
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BGM. INNSBRUCK
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MUÑOZ BOLAÑOS
Juan David
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STAATSANGEHÖRIGKEIT / NATIONALITY
GEBURTSDATUM / DATE OF BIRTH
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16 05 2026
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